Tazrian Noor

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RESEARCH INTERESTS

Optoelectronics, Photonic Integrated Circuits, Optical Computing, Metasurfaces, Metamaterials, Quantum Computing, Machine Learning

EDUCATION

Bachelors of Science in Electrical and Electronic Engineering January, 2020 – December, 2023
Independent University, Bangladesh (IUB), Dhaka, Bangladesh CGPA: 3.67/4.00 (Cum Laude)
Thesis Title: *Investigating the effect of graphene on the opto-electronic performance of thin-film solar cells.*
Supervisor: Dr. Mustafa Habib Chowdhury

International Advanced Levels (IAL) June, 2019
British Council, Sylhet, Bangladesh
Grades: Chemistry (A*), Physics (A), Mathematics (A), Further Pure Mathematics (A)

International General Certificate of Secondary Education (IGCSE) June, 2017
Anandaniketan, Sylhet, Bangladesh
Grades: Chemistry (A*), Physics (A*), Biology (A*), Mathematics (A*), Bangla (A*), Further Pure Mathematics (A), ICT (A), English (A)

RESEARCH EXPERIENCE

Independent University, Bangladesh Dhaka, Bangladesh
Research Scientist November, 2024 – Present
Duties:

- Computationally investigate methods for optimizing lead-free perovskite solar cells using light trapping techniques such as plasmonic nanoparticles.
- Assist PI in drafting research grant applications.
- Mentored six undergraduate research assistants.

Research Assistant March, 2022 – November, 2024
Duties:

- Investigated the effect of graphene and carbon nanotubes on optoelectronic performance of silicon thin-film solar cells.
- Led a study and published a part of the results from the thesis in a peer-reviewed conference.
- Assisted in projects on optimizing cadmium telluride thin-film solar cells using light-trapping techniques such as plasmonics, nanograting and surface texturing.

PUBLICATIONS

Journal Articles (peer reviewed)

[1] R. B. Sultan, A. Al Suny, M. H. Hossain, **T. Noor**, and M. H. Chowdhury, "Particle swarm optimization for performance enhancement of cadmium telluride thin film solar cells by embedded nano-grating structures with a plasmonic metal coating layer," *Heliyon*, vol. 10, no. 19, e38775, 2024, ISSN: 2405-8440. DOI: 10.1016/j.heliyon.2024.e38775.

Preprints

[1] A. A. Suny, **T. Noor**, M. H. Hossain, A. F. M. A. U. Sheikh, and M. H. Chowdhury, "Broadband absorption in cadmium telluride thin-film solar cells via composite light trapping techniques," 2025. arXiv: 2502.20539 [physics.optics]. [Online]. Available: https://arxiv.org/abs/2502.20539.

Conference Proceedings (peer reviewed)

[1] **T. Noor**, F. Mahmud, H. Sharma, A. A. Suny, A. Bushra, and M. H. Chowdhury, "Using plasmonic nanoparticles to enhance the optoelectronic performance of lead-free formamidinium tin halide perovskite solar cells," in *2025 IEEE Region 10 Symposium (TENSYMP)*, (Accepted for presentation), IEEE, 2025.

[2] **T. Noor**, M. H. Hossain, A. Pramanik, A. A. Suny, R. B. Sultan, S. Tohfa, and M. H. Chowdhury, "Enhancing the opto-electronic performance of thin-film solar cells using embedded nanorods comprised of single walled carbon nanotubes," in *2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, 2023, pp. 1–6. DOI: 10.1109/STI59863.2023.10464761.

[3] A. Al Suny, S. Tohfa, R. B. Sultan, M. H. Hossain, **T. Noor**, and M. H. Chowdhury, "Enhancing the opto-electronic performance of Cadmium Telluride thin-film solar cells using nanocone-shaped surface texturing," in *2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, IEEE, 2023, pp. 1–6. DOI: 10.1109/STI59863.2023.10465147.

[4] R. B. Sultan, A. Al Suny, S. Tohfa, **T. Noor**, M. H. Hossain, and M. H. Chowdhury, "Use of nano-grating structures embedded within the absorbing substrate to optimize the efficiency of cadmium telluride thin-film solar cells," in *TENCON 2023-2023 IEEE Region 10 Conference (TENCON)*, IEEE, 2023, pp. 1234–1239. DOI: 10.1109/TENCON58879.2023.10322462.

RESEARCH GRANTS

IUB Office of Sponsored Research Project (Research Project Number: 2024-SETS-11) October, 2024 – October, 2025
Title: *Computational study using SCAPS 1D and FDTD to optimize the optoelectronic performance of multi-junction perovskite thin-film solar cells.*
Principal Investigator: Dr. Mustafa Habib Chowdhury
My Role: Research Scientist

TEACHING EXPERIENCE

Department of Electrical and Electronic Engineering, (IUB) Dhaka, Bangladesh
Teaching Assistant October, 2023 – August, 2024
Course name: Electronics-II (EEE234), Power System Lab (EEE419L), Microprocessor and Interfacing Lab (EEE323L), Communication Engineering Lab (ETE322L)
Duties:

- Conducted tutorial hours based on lectures provided by course instructor.
- Evaluated and provided feedback on in-class assignments and laboratory experiments along with regular weekly consultations.
- Assisted course instructors in invigilating exams.

Department of Physics, (IUB) Dhaka, Bangladesh
Teaching Assistant February, 2023 – May, 2023
Course name: Physics 101 Lab, Physics 102 Lab
Duties:

- Conducted tutorial hours based on lectures provided by course instructor.
- Evaluated and provided feedback on in-class assignments and laboratory experiments along with regular weekly consultations.
- Assisted course instructors in invigilating exams.

AWARDS

Academic Award, Independent University, Bangladesh (IUB) 2020 – 2023

- 100% merit based scholarship for two academic years.
- Vice Chancellor's Honor List (Summer 2021)
- Vice Chancellor's List (Autumn 2021, Spring 2023, Summer 2023, Autumn 2023)
- Dean's Honor List (Spring 2021)
- Dean's Merit List (Autumn 2022)
- Dean's List (Spring 2020, Autumn 2020, Summer 2022)

IUB EEE Innovation Fest 2024 2024

- 2nd Runner Up in Innovative Idea Pitching

Edexcel Higher Achievers' Award, IAL 2020
Daily Star Award, IGCSE 2018
Edexcel Higher Achievers' Award, IGCSE 2017

SKILLS

- **Simulation & Design:** ANSYS LUMERICAL (FDTD, CHARGE), SCAPS 1D, CADENCE (GPDK 90nm), Simulink, PSIM, Proteus, PSpice, AUTOCAD Electrical, Cisco Packet Tracer
- **Programming:** Python, MATLAB, Lumerical Scripting Language, C++, C, Assembly, Verilog, VHDL, Arduino, Markdown, Git
- **Python Libraries:** Pandas, NumPy, Matplotlib, OpenCV, Seaborn, PyGame
- **Virtualization Software:** VMware, VirtualBox
- **Typesetting:** LaTeX
- **Operating Systems:** Windows, Linux (Ubuntu, CentOS)

RESEARCH PRESENTATIONS

[1] *Particle swarm optimization for performance enhancement of cadmium telluride thin film solar cells by embedded nano-grating structures with a plasmonic metal coating layer*, **Poster Presentation** at Idea Pitching and Innovative Project Showcasing, IUB EEE Innovation Fest 2024, 16th October, 2024, Independent University, Bangladesh, Dhaka **(2nd Runner UP)**.

[2] *Enhancing the opto-electronic performance of thin-film solar cells using embedded nanorods comprised of single walled Carbon nanotubes*, **Oral Presentation** at 2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI), 9th-10th December, 2023, Green University, Bangladesh, Dhaka.

[3] *Effect of solar path throughout the day on the opto-electronic performance of plasmonic thin film solar cells*, **Oral Presentation** at IEEE Computer Society Bangladesh Chapter Summer Symposium 2023, 9th-10th June, 2023, Islamic University, Kushtia.

REFERENCES

Available upon request.